# Healthcare IT Fundamentals - Section 2, Lecture 3

# Patient Safety, Clinical Decision Support (CDS) & Medication Error Prevention (Simple Notes)

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# Big Picture

In previous lectures, we learned:

- EHR architecture
- Clinical workflows
- Data integration
- Structured vs unstructured data

This lecture focuses on:

> How EHR systems actively prevent patient harm.

Main topic:

**Using technology to reduce medication errors and adverse events.**

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# 1. The Problem with Paper-Based Healthcare

Before EHRs, healthcare relied heavily on:

- Paper charts
- Handwritten prescriptions
- Human memory

This created many opportunities for mistakes.

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## Example Scenario

Patient visits a specialist.

Doctor writes:

```
Prescribe Drug A
```

Problems:

- Patient forgets to mention allergy
- Doctor doesn't know all medications
- Pharmacist misreads handwriting

Result:

❌ Wrong medication

❌ Allergic reaction

❌ Serious harm

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# Why Paper Systems Are Risky

Paper systems depend on:

```
Doctor Memory
+
Patient Memory
+
Pharmacist Interpretation
```

Humans make mistakes.

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# Swiss Cheese Model

A famous patient safety concept.

Imagine multiple slices of Swiss cheese.

Each slice is a safety layer.

Example layers:

1. Patient
2. Doctor
3. Pharmacist
4. Nurse

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## Problem

Every layer has holes.

```
Layer 1:  O
Layer 2:    O
Layer 3:      O
Layer 4:        O
```

If holes line up:

Error reaches patient.

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## Example

Patient forgets allergy

↓

Doctor misses allergy

↓

Pharmacist misses allergy

↓

Patient receives medication

↓

Adverse reaction

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# 2. How EHR Improves Patient Safety

Modern EHR acts like an automatic safety inspector.

It continuously checks for risks.

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# Example

Doctor enters medication through CPOE.

```
Prescribe Penicillin
```

System instantly checks:

- Allergies
- Existing medications
- Medical history
- Dosage rules

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### If Problem Found

System generates alert.

```
WARNING:
Patient allergic to Penicillin
```

Error caught before reaching patient.

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# Safety Layers in EHR

```
Doctor
   ↓
CPOE
   ↓
Allergy Check
   ↓
Drug Interaction Check
   ↓
Dosage Check
   ↓
Pharmacy
   ↓
Patient
```

Multiple automated defenses.

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# 3. Clinical Decision Support (CDS)

CDS is the intelligence layer of EHR.

Purpose:

Help clinicians make safer decisions.

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# Real-Time Alerts

CDS monitors actions as they occur.

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### Example 1

Drug Allergy Alert

```
Patient allergic to Drug X
```

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### Example 2

Drug Interaction Alert

Patient already taking:

```
Blood Thinner
```

Doctor prescribes:

```
Drug B
```

System detects dangerous interaction.

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### Example 3

Dosage Alert

Doctor enters:

```
5000 mg
```

Normal dosage:

```
500 mg
```

System warns immediately.

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# 4. Soft Stops vs Hard Stops

Not all alerts are equal.

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# Soft Stop

Warning appears.

Doctor can continue after providing reason.

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### Example

```
Warning:
Potential interaction detected.

Continue?
```

Doctor may override.

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# Hard Stop

System blocks the action.

Cannot continue.

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### Example

```
Severe contraindication detected.
Order blocked.
```

Doctor must change order.

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# Easy Memory Trick

| Type | Meaning |
|---|---|
| Soft Stop | Warning |
| Hard Stop | Block |

---

# 5. Evidence-Based Medicine

Healthcare should follow proven medical guidelines.

CDS helps enforce these standards.

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# Problem Without CDS

Doctor must remember:

- Tests
- Medications
- Procedures
- Guidelines

for hundreds of diseases.

Impossible to remember everything.

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# Solution: Order Sets

Predefined treatment templates.

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## Example: Pneumonia

System automatically suggests:

- Chest X-ray
- Blood tests
- Antibiotics

based on current clinical guidelines.

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Instead of:

```
Doctor remembers everything
```

System assists.

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## Benefits

✅ Consistent care

✅ Fewer missed steps

✅ Reduced cognitive load

✅ Better patient outcomes

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# 6. Standardization of Care

Without EHR:

Different doctors may treat same condition differently.

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Example:

Doctor A orders:

```
5 tests
```

Doctor B orders:

```
2 tests
```

Variation increases risk.

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With standardized order sets:

Everyone follows evidence-based workflow.

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Result:

More predictable outcomes.

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# 7. Alert Fatigue

One of the biggest challenges in Healthcare IT.

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# What is Alert Fatigue?

Too many alerts.

Doctors become desensitized.

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Example:

```
Alert
Alert
Alert
Alert
Alert
Alert
```

Eventually:

```
Click Ignore
```

without reading.

---

# Risk

Critical alerts may also be ignored.

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### Real-World Problem

90% of alerts may be low-value.

Doctors learn to dismiss everything.

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# Solution

Organizations must:

- Remove unnecessary alerts
- Prioritize critical warnings
- Monitor override rates

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# 8. Governance Committees

Hospitals cannot randomly modify EHR rules.

Special committees manage:

- CDS rules
- Alerts
- Order sets
- Clinical workflows

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## Responsibilities

Review:

- Alert effectiveness
- Safety incidents
- User feedback

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Goal:

Keep system useful and safe.

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# 9. Downtime Procedures

Technology can fail.

Servers can go down.

Networks can fail.

Power can fail.

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# Important Question

What happens if EHR is unavailable?

Hospitals need:

## Downtime Procedures

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### Example

During outage:

- Paper forms used
- Manual documentation used
- Backup workflows activated

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# Why Important?

Patient care must continue even if systems fail.

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# 10. Role of Administrative Staff

Patient safety is not only the doctor's responsibility.

Everyone contributes.

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Administrative staff must:

- Report hardware issues
- Verify patient information
- Follow downtime procedures

- Protect equipment

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### Example

Broken workstation not reported.

Doctor cannot access records.

Patient care delayed.

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# 11. System Uptime

Healthcare systems must be highly available.

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### Example

Banking downtime:

```
Annoying
```

Hospital downtime:

```
Potentially life-threatening
```

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This is why healthcare systems need:

- Backup servers
- Disaster recovery
- Power backups
- Monitoring

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# 12. User Access Tracking

Every user gets unique credentials.

Never share accounts.

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Why?

Auditability.

System can track:

```
Who
Did What
When
```

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### Example

Medication changed.

System records:

```
User: Dr Smith
Time: 10:05 AM
Action: Medication Updated
```

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Critical for:

- HIPAA compliance
- Investigations
- Patient safety reviews

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# 13. Human Side of Technology

Technology helps.

But technology can also create new problems.

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# Screen-Staring Problem

Doctor focuses on computer.

Patient feels ignored.

### Bad Experience

```
Doctor → Screen
Patient → Waiting
```

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Patient-provider relationship suffers.

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# Triangle of Care

Best practice.

Arrange:

```
Patient
   / \
  /   \
Doctor---Computer
```

All three participate.

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### Benefits

Patient sees:

- Test results
- Charts
- Images

during discussion.

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Technology becomes collaborative.

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# Voice Recognition

Newer EHRs increasingly support:

- Speech-to-text
- Voice dictation
- AI-assisted documentation

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Benefits:

✅ More eye contact

✅ Faster documentation

✅ Less typing

✅ Better patient interaction

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# Key Takeaways

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## EHR Improves Safety

Through:

- CPOE
- CDS
- Allergy checks
- Interaction checks
- Dosage validation

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## Soft Stop

Warning only.

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## Hard Stop

Action blocked.

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## Order Sets

Standardized treatment templates.

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## Alert Fatigue

Too many alerts causing users to ignore them.

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## Governance

Committees manage CDS rules and alerts.

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## Downtime Procedures

Backup workflows when EHR is unavailable.

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## Audit Trails

Track who did what and when.

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## Triangle of Care

Patient + Provider + Computer working together.

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# End of Section 2 Summary

Section 2 taught us:

### EHR Architecture

What components exist.

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### Clinical Workflow

How data flows through the system.

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### Patient Safety

How CDS and CPOE reduce errors.

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### Data Quality

Why structured, accurate documentation matters.

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### Integration

How departments share information.

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### Human Factors

Technology should support care, not replace human interaction.

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# For a Ruby on Rails Developer

Think of CDS as:

```
before_save :validate_patient_safety
```

Before a medication order is committed:

```
check_allergies
check_drug_interactions
check_dosage
check_contraindications
```

If validation fails:

```
  raise PatientSafetyError
```

In simple terms:

**Clinical Decision Support is like a giant real-time validation layer for healthcare, ensuring dangerous actions are caught before they affect the patient.** 🚀